aqualinc
water & soil intelligence

REPORT

PEER REVIEW
Hāwea Basin Groundwater Model

PREPARED FOR
Otago Regional Council

WL24007

22/11/2023

PREPARED BY
Julian Weir

aqualinc.com

Quality Control

Client	Otago Regional Council
Document Title	Peer Review - Hāwea Basin Groundwater Model
Document Number	2
Authors	Julian Weir
Reviewed By	Helen Rutter
Approved By	Helen Rutter
Date Issued	22/11/2023
Project Number	WL24007
Document Status	Final
File Name	Peer Review_Hawea GW Models_Final.docx

For more information regarding this document please contact

Julian Weir
Principal Engineer
Aqualinc Research Limited
(027) 451 5004
j.weir@aqualinc.co.nz

The preferred citation for this document is:

Weir, J.(2023): Peer Review - Hāwea Basin Groundwater Model. Otago Regional Council, Aqualinc Report 2. Aqualinc Research Ltd.

DOCUMENT CONTROL

Version Number	Date	Name
1	09/10/2023	Draft for client review
2	22/11/2023	Final

TABLE OF CONTENTS

EXECUTIVE SUMMARY

Otago Regional Council (ORC) commissioned Lincoln Agritech (LA) to develop a groundwater flow model of the Hāwea Basin aquifer system. This model has been developed to support groundwater allocation decision making that will feed into ORC's proposed Land & Water Regional Plan.

ORC engaged Aqualinc Research Ltd (Aqualinc) to provide an independent peer review of the model. The primary purpose of the peer review is to provide advice on whether it is fit for purpose, but also to consider whether or not the model is defensible and if any recommendations are supported by the findings.

Overall, the Hāwea Basin model has been soundly developed, accounting appropriately for key hydrogeological drivers. The use of a transient model means that effects during dry periods (and potential cumulative effects from year to year) can be quantified (noting that it is often the dry periods that dictate allocation decisions). Conceptualisation and development have been thorough, and model limitations have been transparently discussed. LA have recommended additional studies to improve model representation, and we agree these, along with some additional (relatively minor) recommendations.

We have no hesitation concluding that that model is fit for informing decisions relating to ORC's groundwater allocation limits and zones in the proposed Otago Land & Water Regional Plan.

Numerical models (like this) are a valuable digital asset. They require significant time and cost investments, and if executed well they will provide ORC with a valuable tool for decision making that is sound and defensible. We recommend ORC consider treating numerical models as digital assets, similar to how other Council-owned assets are managed, through a natural lifecycle that includes the scheduling (and funding) of on-going performance monitoring, maintenance and upgrades. This will enable careful and thought-out model development that can be defended, and used soundly to inform policy decisions.

1 INTRODUCTION

The Hāwea Basin groundwater model is located between Lake Hāwea and the Clutha River, Central Otago (Figure 1). The area is semi-arid with low rainfall and high evapotranspiration, particularly during summer periods. Natural recharge to groundwater is low, but the aquifer receives recharge from the lake, from rainfall and irrigation losses (including the Hāwea Irrigation Scheme) and from rivers that flow onto the flats from adjacent hillslopes. The aquifer system largely discharges into the Hāwea and Clutha rivers. Although a relatively small component of the overall water balance, groundwater abstraction has increased over recent years with a compounding scale of effects and consequential competing demand for use.

Otago Regional Council (ORC) commissioned Lincoln Agritech (LA) to develop a groundwater flow model of the aquifer system. It is intended that this model will be used to provide a basis for reviewing groundwater allocation in the area and to inform decisions regarding Otago's proposed Land & Water Regional Plan.

ORC engaged Aqualinc Research Ltd (Aqualinc) to provide an independent peer review of the model. This review has been guided by the Australian Groundwater Modelling Guidelines (Waterlines, 2012)[1]. While this document is now a decade old, its content is still useful, relevant, and provides a robust structure upon which to conduct a model review. New Zealand's own model audit guidelines (PDP, 2002)[2] are somewhat older and are largely obsolete due to advances in contemporary modelling techniques. Relevant aspects of this document are encompassed in the Waterlines (2012) guidelines. It is noted in Waterlines (2012) that these guidelines *'should be seen as a point of reference and not as a rigid standard'.*

The primary purpose of the model peer review is to provide advice on whether the model is fit for purpose. In addition, the review has considered whether or not the model is defensible and if any recommendations are supported by the findings. Key aspects that the review has focussed on are:

- Clarity of the model objectives and whether or not they have been met;
- The conceptual hydrogeology and how this has been represented in the model relative to the objectives;
- Suitability of the software and modelling techniques used to represent the various boundary conditions (including 3rd party software for calculating model inputs like land surface recharge);
- Appropriateness of the level of calibration achieved and the overall water balances;
- Appropriateness of the level of uncertainty analyses undertaken and the scenarios run; and
- Overall completeness of the information presented in the reports.

The following information has been provided by LA for the peer review:

- A report describing the current state of understanding of the groundwater system (including a preliminary conceptual model of the study area) (LA, 2022)[3];
- A report describing the model construction, calibration and scenario results (LA, 2023)[4]
- Digital native MODFLOW files for the model, which we have imported into GMS (2023)[5] for visualising; and
- Access to the GitHub repository (hosted by Komanawa Solutions)[6] where the model and extended documentation is stored.

This report documents the findings of the peer review. Initially, the objectives of the model are noted. Then a brief overview of the model is provided followed by commentary on the other key areas of focus (as noted above). Finally, general comments on the overall completeness of the information presented are listed.

[1] Waterlines (2012): *Australian Groundwater Modelling Guidelines.* Prepared by Sinclair Knights Mers and the National Centre for Groundwater Research and Training for the National Water Commission. Waterlines report Series No. 82, June 2012.
[2] PDP (2002): *Groundwater Model Audit Guidelines.* Prepared for Ministry for the Environment. Pattle Delamore Partners Ltd. October 2002.
[3] LA (2022): *Hāwea Basin Current State Review.* Report 1052-08-01R1, prepared by Lincoln Agritech for Otago Regional Council. 7 April 2022.
[4] LA (2023): *Hāwea Basin Groundwater Model & Allocation Review.* Report prepared by Lincoln Agritech for Otago Regional Council. 4 May 2023.
[5] GMS (2023): *Groundwater Modelling System (version 10.7.6, 64 bit).* Software developed and supported by Aquaveo LLC, USA. Latest build date 14 August 2023.
[6] https://github.com/Komanawa-Solutions-Ltd/Z22031HAW_hawea-model

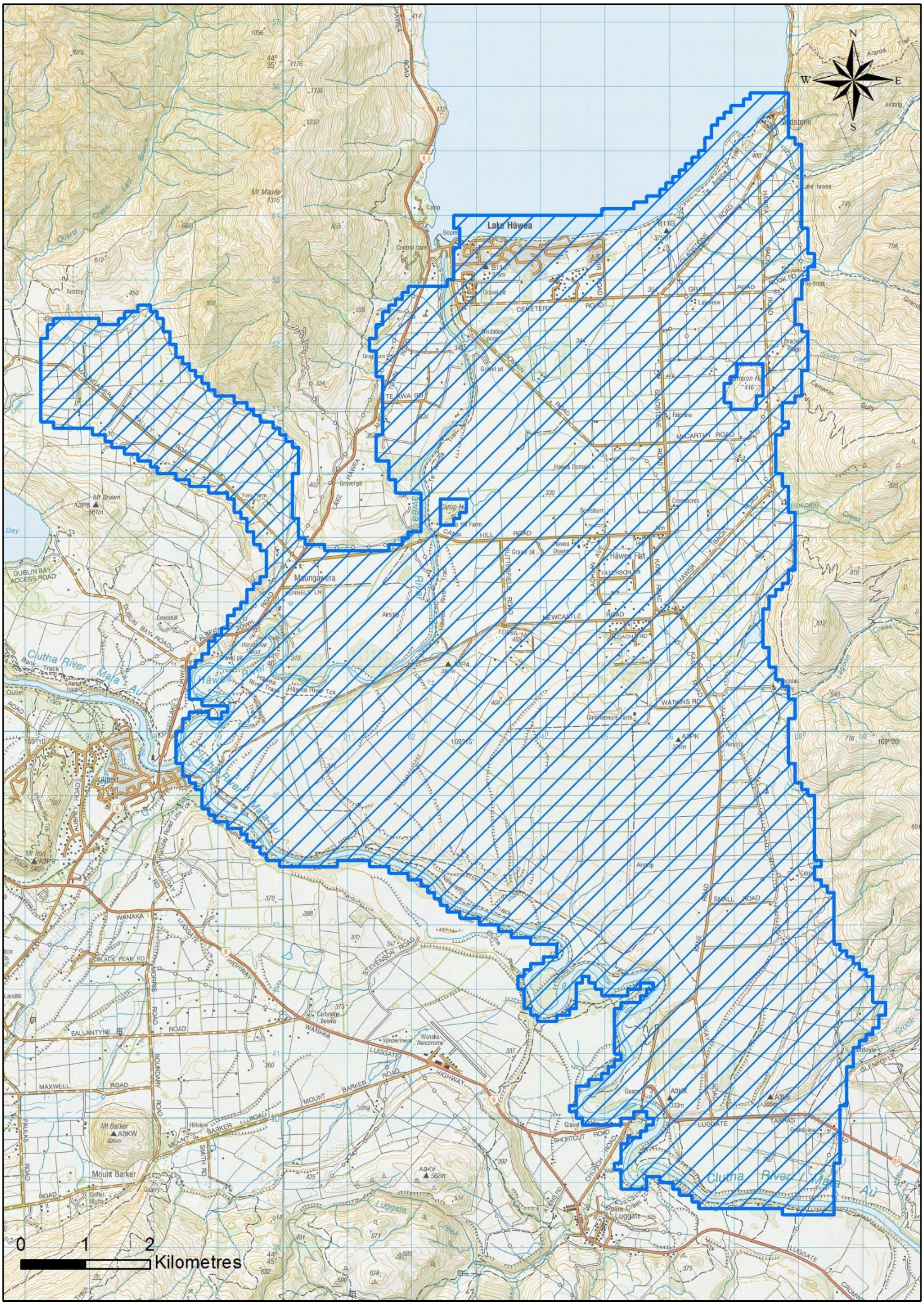

Figure 1: Model domain

2 MODEL OBJECTIVES

In 2012, ORC prepared a hydrogeological review and associated groundwater model of the Hāwea groundwater basin (Wilson, 2012)[7]. This study highlighted data deficiencies that were subsequently actioned by ORC. With new data collected and a greater understanding of the hydrogeology of the area, ORC identified deficiencies of this earlier study[8] including the use of a steady state model to represent seasonal water use (primarily irrigation) and associated allocation recommendations. Key catchment and knowledge changes include:

- Significant changes in land and water use, including increased area (and conversion to) more efficient spray irrigation methods. This has led to some competition for groundwater use rights, plus the availability of new water use records to better quantify actual consumption.
- Expanded residential development and associated water supply[9].
- Confirmation that lake levels are a dominating influence on groundwater levels with the influence reducing at increasing distances from the lake (as expected).
- Improved understanding of land surface recharge processes, including simulation over longer periods to provide a greater range in climate variability.
- The hydrogeological link between groundwater and two significant wetlands.

Therefore, the primary focus of the current study is to:

- Update the Hāwea Basin numerical model with new knowledge gained since the Wilson (2012) study; and
- Review the Wilson (2012) allocation recommendations and provide updated recommendations.

Findings of this study will provide support for the determination of groundwater allocation limits and zones as part of the proposed Otago Land & Water Regional Plan for Otago.

The study excludes consideration of water quality implications.

3 CONCEPTUAL HYDROGEOLOGY

The Hāwea basin groundwater study domain spans an area of approximately 10,800 ha over the Hāwea Flats between Lake Hāwea and the Cutha River. It is bounded by the lake to the north and by the Clutha River to the south. The lower slopes of the Grandview Ranges bound to the east and Mt Maude Range to the east. The study area also includes the off-shoot Maungawera Valley to the north-west as the study concluded that this valley was not hydraulically isolated from the flats aquifer system and contains its own water allocation challenges.

From Aqualinc (2023)[10], the total irrigated area within the model domain is currently approximately 5,130 ha, with irrigation methods primarily pivots, and relatively small areas of border dyke and other spray methods. Irrigation water is supplied from groundwater and from the Hāwea Irrigation Scheme (that sources water from Lake Hāwea) (Figure 2). Pasture is the predominant land use with growing residential and rural-residential zones.

Lake Hāwea provides the majority of inflows to the groundwater system. Recharge is also provided from rivers, rainfall, irrigation, conveyance losses and run-off from bounding hill slopes. Primary groundwater outflows are to surface water (the Clutha and Hāwea rivers) and to abstraction.

[7] Wilson (2012): *Hawea Basin Groundwater Review.* Otago Regional Council Resource Science Unit. October 2012.
[8] Development, monitoring, review and redevelopment is a natural and healthy progression of any hydrogeological study.
[9] Aqualinc assisted Queenstown Lakes District Council with redevelopment of the Lake Hawea well field.
[10] Aqualinc (2023): *Otago irrigated area spatial dataset: 2023 update.* Memorandum prepared for Otago Regional Council. RD23024. Aqualinc Research Ltd. 29 June 2023.

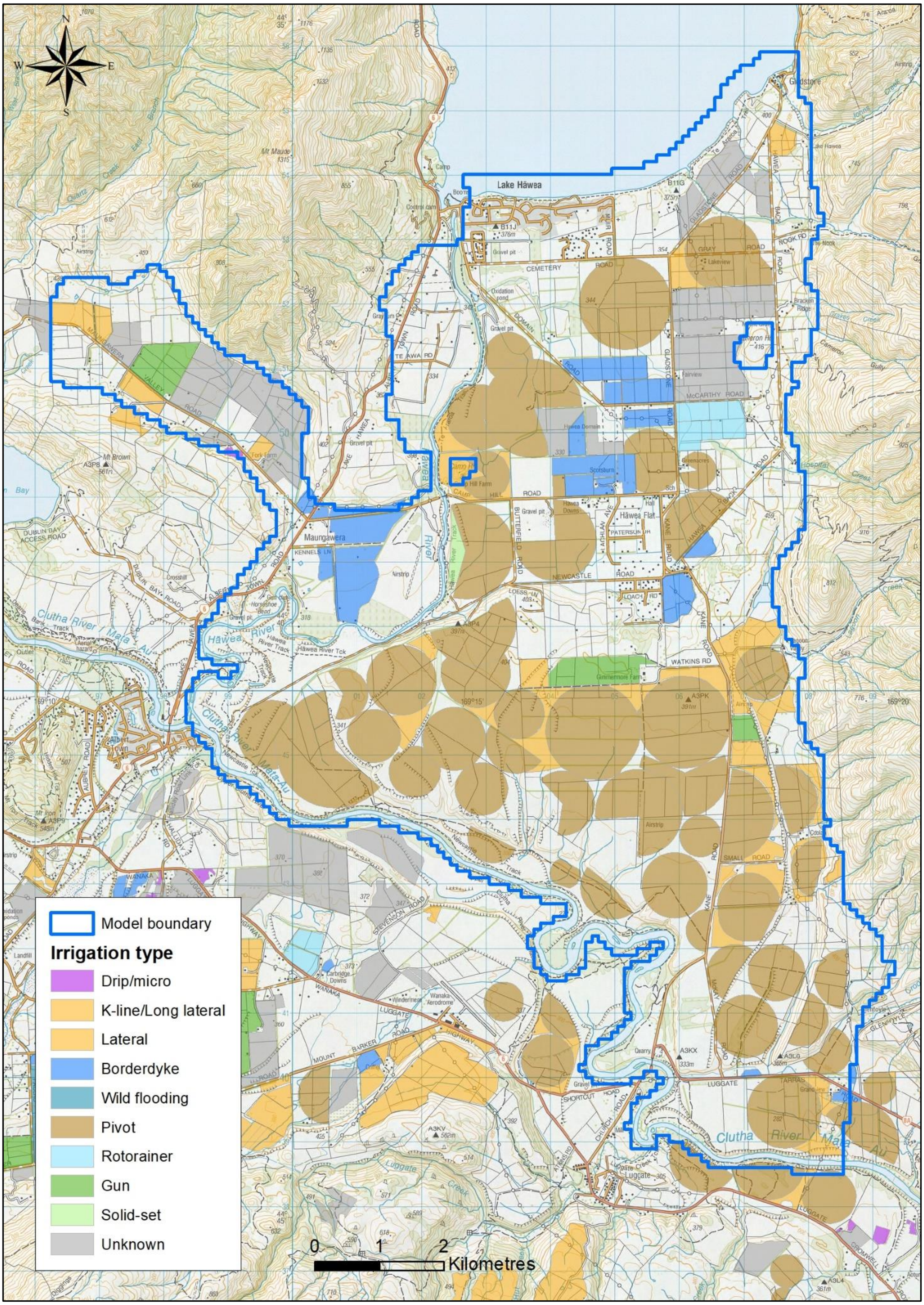

Figure 2: Irrigated areas and methods

From Section 2 of LA (2023), new data has highlighted new understandings of the system, such as:

- Lake-aquifer interactions, including the concept of an underground 'waterfall'. This is not a true waterfall as we might imagine in surface waterways. Rather, it is a sub-surface ridge (or terminal moraine) of low-permeability material over which groundwater flows as it moves from the lake and out to the greater aquifer system. This seems plausible (given Aqualinc's investigations for the Hāwea well field)[11] and additional geological and/or geophysical study could be used to confirm the presence of this feature. The feature has been represented in the model by the use of lower conductivity values (discussed later).
- Propagation of changes in lake levels from the lake and into the wider aquifer system, with timing lags and dampening as the changes travel further into the aquifer system. This is logical and not surprising, and is not dissimilar to the response in groundwater levels from a tidal cycle (though the changes in lake levels are not regular like tides). As a consequence, groundwater levels are highly dependent on lake levels (more so than the effects of changing land surface recharge (LSR) and abstraction). Again, this is logical and it will therefore be important that the transient lake head boundary is well defined.
- The scale and timing of recharge from adjacent hillslopes.
- Further uncertainty on the influence of the Grandview Fault on groundwater flows.
- Greater appreciation of Butterfield and Campbell wetlands and the groundwater catchments that influence theses wetlands.
- The hydraulic response from groundwater abstractions located near rivers.

From the available data, the groundwater model has been adequately conceptualised to meet the study objectives and to accommodate the new hydrogeological understandings noted above. The key hydrogeological features that have been included are:

- Subsurface inflow from Lake Hāwea;
- River boundaries to represent the Clutha and Hāwea rivers;
- No-flow boundaries along the foothills;
- Recharge from smaller rivers and streams that pass onto the flats;
- Land surface recharge from rainfall, irrigation and conveyance losses);
- Groundwater abstraction for irrigation and potable supply;
- A total thickness (which informs the model base) from the earlier Wilson (2012) study that considered drilled boreholes; and
- Two significant wetlands, the responses in which have been derived from underlying groundwater levels.

3.1 Software Suitability and Boundary Conditions

LA have used MODFLOW to simulate the Hāwea Basin aquifer system. This is an industry-standard tool that is appropriate to simulate the study area. They have used MODFLOW-NWT, which is an older, but well tested, version of the code that is in common use throughout the world. A regular 100 x 100 m grid has been applied, which is adequate for the overall scale of the study.

Vertically, the model grid represents a single hydrological layer that has been divided into three numerical layers (of varying thicknesses). This is also standard practice and allows for the simulation of vertical anisotropy should this be needed for calibration. Land surface elevations have been defined from a digital elevation model (DEM). The overall model thickness (which is spatially variable) has been defined based on the thicknesses of the earlier Wilson (2012) study and expanded with new information into areas where the earlier study did not cover. The three-dimensional structure is suitable for the purpose of the model.

[11] Edkins, R (2012): *Lake Hāwea Public Water Supply – Aquifer testing and well field design.* Aqualinc report No. C11036/2, prepared for Queenstown Lakes District Council. February 2012.

3.1.1 Surface Water Packages

MODFLOW's stream (STR) package has been used to represent the Hāwea and Clutha rivers as well as the smaller Grandview and John creeks. This is appropriate and allows for the stream to gain or lose depending on relative height differences between the river and underlying groundwater level at each model cell over which the stream is modelled. The STR package does not allow the direct calculation of flows in the river, but this can be calculated post-model by adding the net stream gains or losses to measured flows at the top modelled reach of the stream. River stage elevations were defined from LiDAR data.

Flows in small creeks that flow onto the flats have been synthesised based on measurements from nearby catchments scaled based on catchment area. This is a sound method to estimate these flows where measured data is not available.

The Butterfield and Campbells wetlands have been demonstrated to be heavily reliant on groundwater levels (as demonstrated in Fig 17 of LA, 2022 for the Campbells wetland). Therefore, these features have not been explicitly represented in the model, but instead their responses have been implied from underlying groundwater levels. This is also appropriate.

3.1.2 Well Package

MODFLOW's well (WEL) package has been used to simulate time-varying groundwater abstraction. Abstraction rates have been derived from ORC's usage data (where available) and processed to fill gaps and extend in time (beyond measured data). This is a sound method to represent historical use. Well depths have primarily been assigned to the top model layer (apart from a few exceptions to avoid the moraine layer and to aid model stability).

The well package has been used (with specified injection rates) to simulate the inputs from smaller streams at each location where they flow from the foothills and onto the flats. This is also appropriate given the relatively low importance of these smaller streams. The method requires that the recharge rate from the streams is calculated independently of the model and then 'hard wired' into the model. This method also assumes that the river does not flow past the location of the specified injection wells (i.e. further out onto the flats) which may not be the case during higher-flow events. However, since allocation is largely focussed on dry periods, this is likely of little consequence.

Near river takes have been excluded from the model (due to cell drying problems). This might suggest that the river-groundwater connections have been assigned inconsistently with actual conditions. Regardless, they are largely river depleting, so if not modelled directly then they can be subtracted from the river flows post-modelling to arrive at a net effect. It is not clear if this has been undertaken.

3.1.3 Land Surface Recharge

The Hāwea area receives land surface recharge rom rainfall and irrigation. Irrigation water is supplied primarily from the Hāwea Irrigation Scheme and also from individual groundwater takes.

Time series of land surface recharge (LSR) have been calculated using the 'Rushton' model used by Wilson (2012). While details of this model are unclear, it appears to include the key features need for a sound soil-water balance model (including an irrigation component and the calculation of actual evapotranspiration from potential evapotranspiration) and inputs have been based on either measured data (where available) or relevant studies (for example, the soil profile available water values in Figure 11 of LA, 2022 are consistent with the latest S-Map data). Therefore, it is likely that this model is appropriate to use. However, additional verification of its suitability to use in the Hāwea Bain area would be helpful (if this exists).

Irrigation race losses have been defined based on the research of McIndoe (2002)[12], which is appropriate given the absence of loss measurements. However, we recommend that additional field work is undertaken to verify this component of groundwater recharge.

[12] McIndoe (2002): *Efficient Irrigation.* Lincoln Environmental report 4521/1, prepared for Environment Canterbury. May 2002.

3.1.4 Parameterisation

The spatial variability of key aquifer hydraulic parameters (hydraulic conductivity and specific yield) have been represented by the use of pilot points. Here, values are specified at given locations (as shown in Figure 10 of LA, 2023) and the values at these points are then interpolated to all cells within the model domain. This is industry-standard practice. Interpolation has been applied using a radial basis function (RBF), which is appropriate.

These pilot points have been assigned over two discrete zones (the high terrace and the remaining study area), plus additional parameters to represent the subsurface moraine zone. This is sensible and demonstrates good use of both hydrogeological knowledge and numerical modelling techniques. Parameter ranges have been assigned adequately large (but not unrealistically) so as to not unduly restrict the calibration process. The parameters allowed to change through the calibration process include hydraulic conductivities, specific storages, specific yields, hillslope recharge multipliers, LSR multipliers, race losses and riverbed conductances.

Results from aquifer tests have been used to constrain the range (upper and lower bounds) of the relevant parameters, which is standard practice to avoid anomalously extreme (high or low) values, though LA acknowledge that some aquifer tests are of low reliability (as reported by Wilson, 2012). A comparison between relevant measured and modelled aquifer test results would be helpful. We also recommend undertaking additional aquifer tests in the study area to verify if the modelled range in hydraulic conductivity is plausible.

Vertical anisotropy has been set to 1:1 (vertical conductivity is the same as horizontal) as vertical gradient are not being modelled. In reality, some vertical anisotropy is likely (e.g. 1:10), but this is of little consequent to the objectives of the study, which are dominated by horizontal flow.

Overall, the model has been soundly parameterised.

4 APPROPRIATENESS OF THE CALIBRATION AND OVERALL WATER BALANCES

Calibration was achieved primarily with the use of PEST (Dohertry, 2010)[13] using single-value decomposition, which is standard industry practice and an appropriate tool to use. The model was calibrated against transient and one-off groundwater level measurements in multiple bores over the study area and against river flows at key sites. Multiple calibration outputs are provided, including hydrographs of groundwater levels and river losses, and residual maps. From these:

- Overall calibration statistics were not found, but these may have simply been located in an area of GitHub that was not discovered. Based on the comparison of measured and modelled groundwater level hydrographs, the model is well calibrated, with dynamic responses well matched.
- Multiple conceptual models were trialled to test various hypothesises (e.g. the lake interface and High Terrace representations), eventually arriving at the final model as presented. This is an excellent modelling technique.
- LA acknowledges that the model does not adequately calibrate to groundwater levels east of the Grandview Fault, and have helpfully proposed an alternative allocation approach for this area.
- Overall average (steady state) flow balances seem sensible given the current understanding of the groundwater system. Groundwater flows are highly dominated by lake recharge, and therefore groundwater levels are highly influenced by lake level.

Mode calibration is excellent and the overall water balances are sensible, given the limited data available for some recharge components.

[13] Doherty, J (2010): *PEST. Model-Independent Parameter Estimation – User Manual. 5th Edition.* Watermark Numerical Computing.

5 APPROPRIATENESS OF THE UNCERTAINTY ANALYSES AND SCENARIOS

5.1 Sensitivity and Uncertainty Analyses

Section 6 of LA (2023) helpfully lists the model limitations and recommendations for additional data collection. The transparency of this section is refreshing, and we agree with the recommendations for further data collection (plus subsequent model review and updating).

A key aspect worthy of note is that the study has not undergone thorough parameter or predictive uncertainty analyses (to quantify the model non-uniqueness). Therefore, review of model uncertainty can't be completed, other than to agree with LA that this should be undertaken as part of good modelling practice.

5.2 Scenarios and Revised Allocation Limits

A range of scenarios has been tested that consider:

- The calibrated parameters and how these affect predictions;
- Low lake levels;
- The sources of water (using MT3D);
- Various allocations from a range of management zones; and
- Wetlands setbacks.

These provide a thorough and sensible range of scenarios that are appropriate to inform ORC's plan change process. The scope of the model review is not sufficient to review each model scenario, but only to conclude that the model is suitable for running these scenarios. LA's definition of adequate well penetration depth[14] is also a helpful criteria to apply.

6 OVERALL COMPLETENESS OF THE INFORMATION PRESENTED

Overall, the Hāwea Basin model has been soundly developed and presented. While the language used in the reports is occasionally overly complex, the reports are (refreshingly) thoroughly written and well structed to enable peer review.

The GitHub repository is a modern tool to manage, process and transparently display model development and associated data, although it requires an understanding of the system and a good understanding of coding (if it is to be used for further model development). The use of this system has been excluded from the peer review process, which has instead focussed on the model itself (rather than the repository). However, some GitHub references have been explored for more complete descriptions of certain components of the model development, which has been helpful.

[14] Their key threshold for evaluating sustainable groundwater use is 1 or more adequate penetration levels, defined as 3 x the seasonal groundwater level variation below the mean.

7 SUMMARY AND CLOSING COMMENTS

Overall, the Hāwea Basin model has been soundly developed, accounting appropriately for key hydrogeological drivers. The use of a transient model means that effects during dry periods (and potential cumulative effects from year to year) can be quantified (noting that it is often the dry periods that dictate allocation decisions). Conceptualisation and development have been thorough, and model limitations have been transparently discussed. We support LA's conclusions on additional data collection to improve model representation. Although relatively minor, the following additional areas could be targeted to feed into future model development.

- Further verification that the 'Rushton' soil-water balance model is appropriate to apply to the Hāwea Basin area;
- Additional field work to verify irrigation scheme race losses to ground;
- Targeted aquifer testing, and a comparison between measured and modelled aquifer test results (where relevant); and
- Undertaking thorough parameter and predictive uncertainty analyses.

We have no hesitation concluding that that model is fit for informing decisions relating to ORC's groundwater allocation limits and zones in the proposed Otago Land & Water Regional Plan. Undertaking some of the above recommendations would add further confidence to the model conclusions.

7.1 Asset Management

Numerical models (like this) are a valuable digital asset. They require significant time and cost investments, and if executed well they will provide ORC with a valuable tool for decision making that is sound and defensible. As noted in Weir (2016)[15], numerical models suffer from natural deterioration due to moving quality benchmarks and expectations, such as:

- New data becoming available (people remember more recent years the best);
- Software becoming superseded and obsolete;
- New modelling techniques becoming the norm; and
- Model outputs changing focus.

Given this, it is worthwhile considering treating numerical models as digital assets, similar to how other Council-owned assets are managed, through a natural lifecycle that includes the scheduling (and funding) of on-going performance monitoring, maintenance and upgrades. This would then avoid a last-minute scramble when the model is needed urgently (say for use in a hearing or important decision making process). It will allow careful and thought-out model development that can be defended.

[15] Weir (2016): *Using Best-Available Data for Robust Groundwater Model Development: An Example.* Abstract for the New Zealand Hydrological Society Conference, Queenstown, November 2016; p 143.